PREDICT-THEN-OPTIMIZE FRAMEWORK FOR PUBLIC TRANSPORT LINE REDESIGN UNDER FLUCTUATING TRAFFIC CONDITIONS

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Abstract: Public Transport (PT) lines are traditionally designed to optimize performance under nominal traffic conditions.zg ~~In practice, however, operating conditions frequently deviate from the nominal ones, leading to substantial performance deterioration. Existing adaptation mechanisms typically rely on corrective and reactive interventions, such as stop-skipping, which prove insufficient under large or recurrent traffic fluctuations. In such contexts, incremental adjustments may be insufficient. This paper evaluates the potential of deeper structural *redesigns* to preserve performance. To this aim, this paper proposes a method to proactively redesign appropriate parts of PT networks when faced with high traffic fluctuations that would otherwise deteriorate operator- and user- performance.~~ We adopt a predict-then-optimize paradigm in which PT lines are reconfigured based on traffic forecasts via an algorithm based on the Non-dominated Sorting Genetic Algorithm III (NSGA-III).zg ~~For operational feasibility, to avoid excessive structural changes, we enforce high Jaccard edge overlap between the original and redesigned networks. To account for the effect of prediction inaccuracies on redesign quality, we construct a statistical model of the errors made by a well-established deep learning-based prediction model (Diffusion Convolutional Recurrent Neural Network) trained on real-world data.~~ zg~~Computational results on Mandl's benchmark network and on the large-scale Beijing network show that the controlled redesign of PT lines brings significant user-centric performance improvement and operational cost reductions in high traffic fluctuation scenarios, while limiting topological modifications.~~Computational results on Mandl's and Beijing networks show improvements while limiting topological modifications. On the Beijing network under high variability, average travel time improves by up to 25.8% while preserving over 85% line overlap; unlike stop-skipping baselines, which break connectivity for a large share of zg~~OD~~origin–destination (OD) pairs, our redesign preserves full OD connectivity.zg ~~This indicates the potential of transitioning from current static planning paradigms toward continuous and adaptive PT network design.~~

Keywords: Public transport network redesign; Predict-then-optimize; Multi-objective optimization; Travel time uncertainty; Structural continuity

1 Introduction

Public Transport (PT) networks are typically planned for nominal, scheduled conditions (Ibarra-Rojas et al., 2015), while real-world operations are continuously subject to variability caused by congestion and other factors (Büchel & Corman, 2020; Danaher et al., 2020), especially in mixed traffic environments. Such routine perturbations occur daily and can propagate across the network, progressively degrading service reliability (Marra & Corman, 2020).zg ~~For example, in Xi'an, 74.34% of bus runs were delayed. Beyond operational impacts, insufficiently reliable public transport can also induce behavioral responses, including increased reliance on private vehicles.~~

zg ~~Existing approaches to cope with recurrent perturbations can be broadly classified as *preventive*, *reactive*, or *proactive*, depending on the decision horizon and intervention level.~~ Preventive strategies enhance robustness ex ante through structural measures, such as dedicated bus lanes that isolate buses from mixed traffic (Russo et al., 2022). However, limited street space and high infrastructure costs often make such measures infeasible in dense urban areas (J.-Q. Li et al., 2009), and preventive capacity tends to be inflexible and underutilizedzg ~~(ITF, 2024, 2025)~~. Reactive strategies attempt, at the operational level, to restore service after perturbations have occurred, for instance via stop-skipping, holding, or detouring (Peled et al., 2021; Tang et al., 2018)zg~~; Zheng et al. (2023)~~.zg ~~These actions deteriorate under high uncertainty, inconvenience passengers, and are insufficient to manage cascading perturbations when no alternative network structure is available.~~ A key limitation shared by both preventive and reactive strategies is that the PT network structure typically remains fixed, despite persistent changes in operating conditions. Network design decisions are revised infrequently, and traditional models assume deterministic or fixed travel times (Büchel & Corman, 2020; Fonseca, 2020), limiting their ability to address recurrent traffic fluctuations.zg ~~Proactive approaches aim to overcome this limitation by anticipating variability through short-term predictions and revising the line design before the next operating interval.~~

The **main contribution** of this paper is to assess the potential of predict-then-optimize proactive redesign of bus networks. To this aim, we propose a new method that adapts bus lines to time-varying stochastic traffic congestion. We solve a multi-objective problem to minimize operational cost and user travel time, and to limit structural network modifications, so as to ease operations and user routing. Adaptation is triggered only when forecasted conditions severely deviate from the nominal ones; moreover, such adaptation is anticipated (e.g., 1h in advance) to avoid disrupting journeys of current travelers. The experiments in this paper, however, evaluate the redesign step itself: namely, the quality of a single forecast-conditioned redesign under varying travel-time variability. zg~~Results on the Mandl benchmark and Beijing networks show that notable improvements in user- and operator-centric performance can be achieved while limiting the extent of structural modifications. In the Beijing large scale network, beyond a one-off static re-optimization gain of 11.4%, with no link travel time variability, the benefits grow with such a variability. When it is very high, average travel time improves by up to 25.8% and operational cost by up to 22.6% at with high traffic variability, while line overlap stays above 85% (Table 5). Compared with a stop-skipping baseline, which improves travel time by at most 19.7% but only by disconnecting 28–39% of OD pairs under high variability, our redesign attains a larger gain while keeping every OD pair reachable. These findings highlight the potential of proactive bus-line redesign in future public transport systems, where adaptive operational strategies may become easier to deploy thanks to vehicle automation and to users' growing familiarity with flexible, app-mediated mobility services, as illustrated by the diffusion of on-demand platforms.~~

2 zg~~Model and Problem Formulation~~ Methodology

2.1 zgPredict-then-optimize framework

zg~~As illustrated in Fig. 1, our methodology follows a predict-then-optimize framework that explicitly integrates current observations into prediction.~~Our methodology follows a predict-then-optimize framework that explicitly integrates current observations into prediction. The prediction model is first trained offline on historical data to learn travel times on road links. During operation, it receives current observations from the latest time steps and outputs a short-term prediction of link travel times $\hat{t}_{ij}$, which are then supplied to the optimization model for potential network redesign. The resulting network design is then evaluated against realized travel times t_{ij} to assess the actual performance of our adaptive network under realistic conditions. To ensure timely responsiveness, this procedure is repeated periodically, i.e., once per hour. At the start of each time slot, the new network is announced in advance (e.g., 1h), so passengers can plan a route that remains valid for their entire trip and is never changed mid-journey; we assume here that the longest trip is shorter than one time slot (1h).zg ~~This also keeps the bus routes easier to manage.~~

2.2 zgGraph Model of PT Network — Network design and objectives

We represent the underlying road infrastructure as an undirected substrate network $\mathcal{G}^{\text{sub}} = (\mathcal{V}^{\text{sub}}, \mathcal{E}^{\text{sub}})$, where $\mathcal{V}^{\text{sub}}$ are candidate bus stops and $\mathcal{E}^{\text{sub}} = \mathcal{V}^{\text{sub}} \times \mathcal{V}^{\text{sub}}$ the potential connections among them. Observe that a single $\ell \in \mathcal{E}^{\text{sub}}$ may represent a sequence of road links from a stop to another. Let $\mathcal{L}$ denote the set of all PT lines in the network. Each PT line $\ell \in \mathcal{L}$ is defined by a set of undirected substrate edges, representing the physical stop-to-stop connections served by the line. To ensure path continuity and avoid disconnected or cyclic line configurations, the undirected edge set of each line is represented through a directed simple path graph $\mathcal{G}_\ell = (\mathcal{V}_\ell, \mathcal{E}_\ell)$ on the undirected substrate $\mathcal{G}^{\text{sub}}$zg~~, where $\mathcal{V}_\ell \subseteq \mathcal{V}^{\text{sub}}$ is the set of stops served by line ℓ, and $\mathcal{E}_\ell \subseteq \mathcal{E}^{\text{sub}}$ denotes the set of connections between consecutive stops along the path representation of line ℓ~~. zgThe model requires all considered origin–destination (OD) pairs to remain connected.

Vehicles operating on line ℓ are assumed to traverse this path in both directions; the reverse traversal is implicitly captured through symmetric travel times (Schmid, 2014). zgIf t^ℓ is its one-way travel time and n^ℓ vehicles are assigned, its headway is $h^\ell = 2t^\ell/n^\ell$. Assuming random passenger arrivals, the average waiting time for line ℓ is given by $w^\ell = h^\ell/2$. zgA transfer to another line adds that line's expected waiting time and a fixed five-minute penalty. The virtual graph makes these boarding and transfer costs explicit when evaluating passenger paths.

We formulate our public transport network adaptation problem as a three-objective optimization, as expressed in (1). The first objective, Z_1, minimizes passenger travel time deterioration, where q_k represents the travel demand flow. The use of $\max(\hat{t}_k - t_k^0, 0)$ guarantees that only travel-time increases relative to the nominal benchmark are penalized, thereby avoiding situations where improvements for certain OD pairs compensate for deterioration in others. The second objective Z_2 minimizes operational costs through total vehicle-hours. The third objective Z_3 minimizes network structural changes using the Jaccard index, calculated as the ratio between the number of shared edges (i.e., those present both in the original and in the re-designed network) and the total number of edges (i.e., the union of the edges in the two networks). Structural continuity is handled as an *objective* (Z_3). zgStructural continuity means keeping the redesigned lines similar to the lines already in service, so that passengers and operators need to adapt to fewer route changes. For example, if the original and redesigned networks each have 10 line-labelled links and share 8, their union has 12 links: the Jaccard overlap is $8/12$, and $Z_3 = 1 - 8/12$. A smaller Z_3 means greater overlap and proportionally less change. It is an objective, not a fixed minimum-overlap constraint, so its trade-off with travel time and cost is shown by the Pareto solutions.

zg~~$\min(Z_1, Z_2, Z_3) = \left(\sum_k q_k \max(\hat{t}_k - t_k^0, 0),\ \sum_\ell n^\ell 2\hat{t}^\ell,\ 1 - \frac{\sum_{\ell,(i,j)} \min(e_{ij}^\ell, e_{ij}^{\ell,0})}{\sum_{\ell,(i,j)} \max(e_{ij}^\ell, e_{ij}^{\ell,0})} \right)$~~

$$\min(Z_1, Z_2, Z_3) = \left(\sum_k q_k \max(\hat{t}_k - t_k^0, 0),\ \sum_\ell n^\ell 2\hat{t}^\ell,\ 1 - \frac{\sum_{\ell,\{i,j\}} \min(x_{ij}^\ell, x_{ij}^{\ell,0})}{\sum_{\ell,\{i,j\}} \max(x_{ij}^\ell, x_{ij}^{\ell,0})} \right). \tag{1}$$

zg~~In Eq. (1), $e_{ij}^{\ell,0}$ and e_{ij}^ℓ denote whether line ℓ uses substrate link (i,j) in the original and the redesigned network, respectively.~~Here $\hat{t}_k$ is the predicted shortest travel time for OD pair k on the redesigned network; t_k^0 is its travel time on the original network under nominal link times. For the Jaccard term, x_{ij}^ℓ denotes use of physical link $\{i,j\}$ by line ℓ, and $x_{ij}^{\ell,0}$ its original value. Each line-labelled physical edge is counted once, irrespective of path orientation. zgHere n^ℓ is the vehicle allocation and $\hat{t}^\ell$ the predicted one-way line travel time; Z_2 is a design-level vehicle-time proxy. Observe that when solving our optimization problem, the realized link travel times t_{ij} are not available, we will thus optimize the decision variables based on prediction $\hat{t}_{ij}$ and then evaluate, ex-post in the results, the objective function evaluated using the realized link travel times t_{ij}.

We now report the constraints that, together with the objective (1), formalize the upper-level network design problem. Passenger route variables are treated as induced (follower) variables and are discussed separately. $\mathcal{N}$: Set of nodes, indexed by $i, j \in \mathcal{N}$; $\mathcal{E}$: Set of feasible edges in the network; y^ℓ: 1 if line ℓ is activated (operated); e_{ij}^ℓ: 1 if edge (i,j) is used by line ℓ in new network; a_i^ℓ, b_i^ℓ: 1 if node i is the

start/end terminal of line ℓ; n^ℓ: Number of vehicles on line ℓ (fleet size); u_i^ℓ: Auxiliary variable for subtour elimination (MTZ): it denotes the order in which stop i appears in line ℓ. zgVariables y, e, a, b are binary, n, u integer, and $\bar{n}$ is the per-line vehicle upper bound.

Line Structure Constraints

$$\sum_{i\in\mathcal{N}} a_i^\ell = y^\ell, \qquad \sum_{i\in\mathcal{N}} b_i^\ell = y^\ell \qquad \forall \ell \in \mathcal{L} \tag{2}$$

$$a_i^\ell + b_i^\ell \le 1 \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{3}$$

$$e_{ij}^\ell + e_{ji}^\ell \le y^\ell \qquad \forall \{i,j\} \in \mathcal{E},\ \forall \ell \in \mathcal{L} \tag{4}$$

$$\sum_{j:\{i,j\}\in\mathcal{E}} e_{ji}^\ell + a_i^\ell = \sum_{j:\{i,j\}\in\mathcal{E}} e_{ij}^\ell + b_i^\ell \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{5}$$

$$\sum_{j:\{i,j\}\in\mathcal{E}} e_{ij}^\ell \le 1, \qquad \sum_{j:\{i,j\}\in\mathcal{E}} e_{ji}^\ell \le 1 \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{6}$$

$$u_i^\ell - u_j^\ell + 1 \le |\mathcal{N}|\left(1 - e_{ij}^\ell\right) \qquad \forall \{i,j\} \in \mathcal{E},\ \forall \ell \in \mathcal{L} \tag{7}$$

$$1 \le u_i^\ell \le |\mathcal{N}| \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{8}$$

$$n^\ell \ge y^\ell, \qquad n^\ell \le \bar{n}\, y^\ell \qquad \forall \ell \in \mathcal{L} \tag{9}$$

$$\sum_{\ell\in\mathcal{L}} n^\ell \le V_{\max} \tag{10}$$

Constraints (2)–(10) establish the topological structure and operational limits of the designed network. Constraints (2)–(3) synchronize line activation with terminal selection, mandating that every active line ($y^\ell = 1$) connects exactly one initial terminal to one final terminal, thereby preventing zero-length loops. Bus route continuity and linearity are enforced by the flow conservation in (5) combined with the degree Constraints in (6); the latter restrict nodes to a maximum of one predecessor and one successor, strictly imposing a non-branching simple path topology. Acyclic integrity is further guaranteed by Constraints (7)–(8), which adapt the Miller-Tucker-Zemlin (MTZ) formulation to eliminate disconnected subtours that might otherwise satisfy flow balancezg~~: if link (i,j) is used in line ℓ, then node j should appear after i along the line (its order of appearance (u_j) must be higher than i)~~. zgThe ordering inequality applies to both candidate directions. For each physical link, $x_{ij}^\ell = e_{ij}^\ell + e_{ji}^\ell$; the original-network value is defined analogously. Finally, resource allocation is governed by Constraints (9)–(10), which assign vehicles only to active lines and enforce the global fleet budget.

To model passenger routing, we construct an auxiliary directed graph $\mathcal{G}^{\text{PAX}} = (\mathcal{V}^{\text{PAX}}, \mathcal{E}^{\text{PAX}})$, referred to as the virtual-node graph. The node set $\mathcal{V}^{\text{PAX}}$ comprises physical stop nodes $i \in \mathcal{V}^{\text{sub}}$ and line-specific virtual nodes v_i^ℓ for each line ℓ serving stop i. zgIts arcs represent in-vehicle travel, access and transfers; shortest paths yield OD travel times. The model assumes uncongested passenger assignment: it does not model crowding or passenger-flow-dependent travel times.

2.3 Solution algorithm

Given the tri-objective nature of the problem and the nonconvex trade-offs, we adopt an NSGA-III frameworkzg ~~(Algorithm 1)~~ (Deb & Jain, 2014) to approximate the Pareto front. The evolutionary search operates on network design variables (line topologies e_{ij}^ℓ and fleet allocations n^ℓ). Each individual encodes a complete PT network design, comprising a set of line path representations and the associated vehicle allocations. zgLike genes in a chromosome, each line and its assigned vehicles form a block: exchanging or modifying these blocks explores different network designs. NSGA-III can search these discrete designs while retaining diverse trade-offs among the three objectives. At each generation, offspring are produced through selection, variation, and evaluation, followed by the NSGA-III reference-point-based *environmental selection*, which prioritizes non-dominated solutions while maintaining diversity along the Pareto front. zgFeasibility-oriented repair addresses line-level structural infeasibility. For each candidate,

shortest paths on a virtual-node graph incorporate in-vehicle and waiting times and provide the OD travel times used in the objectives.

PT lines are either constructed from scratch or obtained by locally perturbing routes reconstructed from the existing network. zgCandidates are screened for simple-path validity, connectivity of the considered OD pairs, and a minimum initial Jaccard similarity $J_{\min} = 0.5$. This initialization threshold should not be confused with an enforced overlap threshold for every subsequent solution: structural continuity is an optimization objective. The crossover exchanges complete lines between parents as the primary recombination mechanism; any subsequent modifications are limited to feasibility restoration. Whenever a line topology is inherited from a parent, its vehicle allocation is inherited jointly, thereby preserving the coupling between bus route geometry and service frequency. With probability P_m, a mutation perturbs a single line ℓ via local structural modifications, including node insertion, node deletion, or segment reconnection. This correction is a feasibility projection and is not guided by objective improvement. zgThe experiments use crossover probability 0.85, mutation probability 0.35, and 125 reference points. The algorithm returns non-dominated alternatives rather than a single universally optimal design.

3 Evaluation method

Our optimization uses predicted link travel times $\hat{t}_{ij}$, while actual link travel times t_{ij} are unknown at decision time. We therefore evaluate whether the generated line plans remain effective even in presence of inevitable prediction errors. We use the Diffusion Convolutional Recurrent Neural Network (DCRNN) (Y. Li et al., 2017) as a representative spatio-temporal predictor to generate empirical one-hour-ahead prediction errors. The objective is not to benchmark forecasting models, but to expose the line-plan generation method to realistic forecast errors. zgLog-relative errors $\eta_{ij} = \log(\hat{t}_{ij}/t_{ij})$ are sampled with replacement from empirical errors; simulated forecasts are $\hat{t}_{ij} = t_{ij}\exp(\eta_{ij})$. The shape we choose for this random variable is a log-normal distribution, which effectively captures the skewed nature of travel time data (Guessous et al., 2014) and accounts for extreme delay scenarios that can occur in reality, although not often. zgFor Mandl, the median is fixed at the nominal link time as cv changes. We use the median rather than the mean to retain a typical nominal time; the mean consequently increases with cv.

Due to the absence of historical traffic data for Mandl's Swiss network, the DCRNN is trained on the METR-LA dataset (Y. Li et al., 2017), which contains traffic speed measurements from 207 sensors in Los Angeles County collected at 5-minute intervals over four months. For the real-world Beijing case study, instead, we utilize the Q-Traffic dataset (Liao et al., 2018), which provides traffic speed and road network data for the 6th Ring Road in Beijing, China. Errors are not differentiated based on the links where they were calculated; consequently, we do not represent link-specific heterogeneity or spatio-temporal correlation.

Mandl's Swiss network, initially used by Mandl (1980), consisting of 15 nodes, 21 links, and a symmetric demand of 15,570 passenger trips, is employed as a canonical benchmark for comparison with existing PT network design methods (Ahern et al., 2022; Liang et al., 2025). To further assess the method's applicability under realistic operating conditions, we consider a real-world urban public transport network in a region in Beijing, China, recently analyzed by Wu et al. (2024), which comprises 301 bus stops, 3,758 road sections, and 29 bus lines. Its origin–destination demand matrix is constructed from smart card data collected during peak hours by aggregating trips between boarding station i and alighting station j. Key parameters are set as follows: a fixed dwell time of $d = 1$ min per stop is assumed following Chen et al. (2022). For Mandl's network, the maximum fleet size ($V_{\max}$) is capped at 40, following the service-frequency setting of Ahern et al. (2022) to maintain comparable headways. For the Beijing network, $V_{\max}$ is set to 150 to accommodate a constant 12-minute headway across this large PT system. zgWe consider coefficients of variation $cv = 0, 0.05, 0.25, 0.50, 1.00$, where cv is the standard deviation divided by the mean; the Mandl stochastic experiments use 30 realizations of link travel times and prediction errors. The optimization uses predicted times, and designs are evaluated using realized times. A compromise solution is selected from the Pareto front by its Euclidean distance to the component-wise median objective vector.

zgThe Mandl experiments compare redesigned networks with static route-frequency solutions and a frequency-only adjustment, in which line structures remain fixed while vehicle allocation changes. The Beijing comparison uses the original network and stop-skipping variants. A one-hour maximum computational time per instance is used for the offline experiments; this is a computational budget, not evidence that a complete operational deployment can be carried out within one hour. zgWe use three indicators. IS_{20} is the percentage of OD pairs whose travel time is at least 20% lower than in the reference network; each OD pair counts equally. OCR is the percentage reduction in the operating-cost proxy $\sum_\ell n^\ell 2t^\ell$ relative to the reference network; positive values indicate savings. For both indicators, the redesigned and reference networks are evaluated under the same realized link times. LO is the number of shared line-labelled edges divided by the number of distinct line-labelled edges in at least one network, expressed as a percentage: 100% means no edge change.

4 Numerical results

4.1 zg~~Main performance results on Mandl's benchmark~~ Mandl benchmark

We use predicted travel times $\hat{t}_{ij}$ as input to the optimization problem, and we evaluate the resulting adaptive network against realized link travel times and prediction errors across multiple scenarios. As travel-time variability increases, both 20% Improvement Share (IS_{20}) and Operational Cost Reduction (OCR) rise systematically, reflecting the increasing value of adaptive redesign under highly fluctuating conditions. Across all variability levels and redesign schemes, Line Overlap (LO) remains high, indicating that these improvements result from targeted adjustments rather than wholesale restructuring. zgTable 1 compares redesigns of four to twelve lines with the static route-frequency solutions of Ahern et al. (2022). For the four-line instance, at $cv = 0$, redesign achieves $IS_{20} = 5.79 \pm 2.99\%$, OCR $= 10.20 \pm 5.31\%$ and LO $= 84.34 \pm 5.44\%$. At $cv = 1$, these become $30.57 \pm 11.60\%$, $39.20 \pm 12.94\%$ and $77.26 \pm 5.19\%$, respectively. Thus, against Ahern et al., more OD pairs gain at least 20% as variability increases, while the reported redesigns retain substantial line overlap.

zg~~The frequency-only baseline $\mathcal{G}_{\text{PT}}^{\text{freq}}$ improves service under variability, but it captures substantially less of the OD-level improvement than full redesign; for example, at $cv = 1$, its 20%-improvement share is roughly half that of our method.~~The frequency-only baseline improves service under variability, but it captures substantially less of the OD-level improvement than full redesign; for example, at $cv = 1$, its 20%-improvement share is $16.47 \pm 9.53\%$, compared with $30.57 \pm 11.60\%$ for our method. zg~~The results suggest topology change is beneficial when congestion patterns create larger spatial mismatches in the original network.~~ While frequency-only adjustment keeps the line topology fixed, the proposed line redesign can reroute service away from deteriorated links. zgIn a forecast-error sensitivity analysis, realized times are held fixed while forecasts $\hat{t}_{ij}(\alpha) = t_{ij}\exp(\alpha\eta_{ij})$ vary with the error scale α: zero gives a perfect forecast, one reproduces empirical error, and larger values stress the model. The hypervolume results indicate that the opportunity for redesign grows with variability and that the empirical-error forecast retains nearly all available forecast value, with uncertainty across realizations.

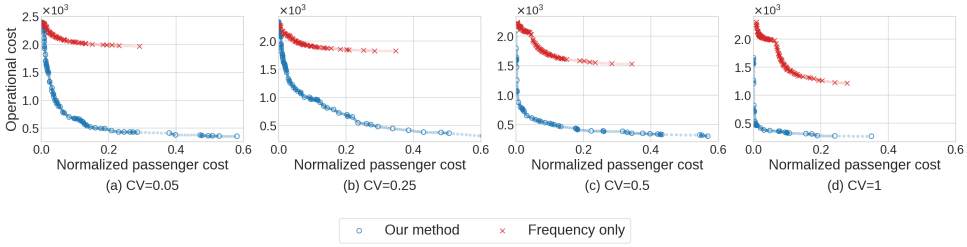


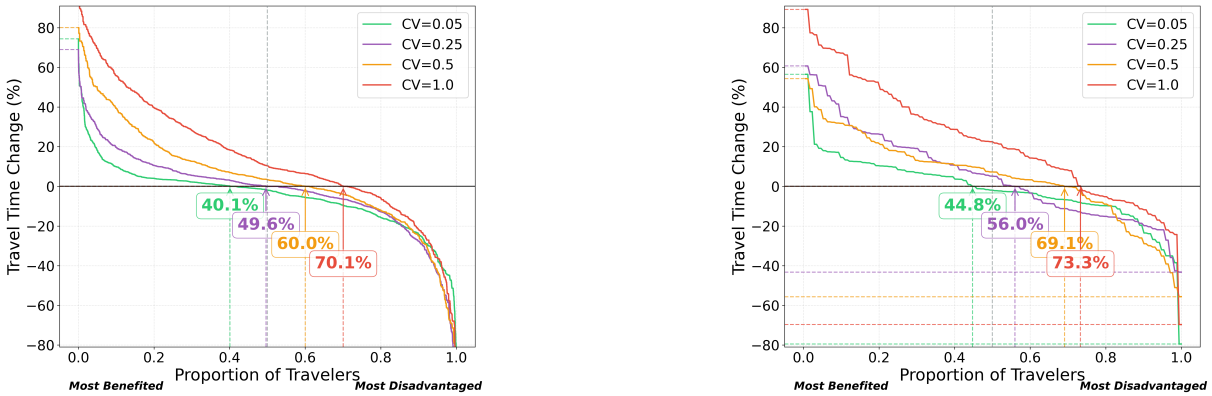
Figure 1: zgTravel-time, cost (Z_1, Z_2) Pareto fronts for the 4-line Mandl instance under four travel-time variability levels.

Fig. 1 plots the 2D projection of the Pareto front, highlighting the trade-off between passenger travel-time Z_1 (which in the figure is normalized between 0 and 1) and operator cost in vehicle-hours (Z_2). Each point maps to a specific network design, with the lower-left corner representing the ideal direction for both

Table 1: Performance comparison of the solutions from Ahern et al. (2022) under different variability levels. zgValues are mean ± standard deviation over 30 realizations; CV is the coefficient of variation.

Lines	Metric	CV=0	CV=0.05	CV=0.25	CV=0.50	CV=1.00
4	IS_{20} (%)	5.79 ±2.99	8.25 ±4.65	15.40 ±5.66	24.63 ±7.91	30.57 ±11.60
	OCR (%)	10.20 ±5.31	13.28 ±7.37	16.86 ±6.74	24.06 ±12.01	39.20 ±12.94
	LO (%)	84.34 ±5.44	83.79 ±7.73	78.83 ±6.65	77.53 ±5.27	77.26 ±5.19
6	IS_{20} (%)	8.96 ±4.47	9.53 ±3.60	16.31 ±5.10	23.46 ±7.84	31.50 ±10.46
	OCR (%)	25.94 ±4.71	28.94 ±5.40	32.10 ±5.69	36.75 ±9.55	48.90 ±10.19
	LO (%)	79.58 ±2.29	81.20 ±2.80	78.16 ±3.94	79.63 ±4.47	79.16 ±4.08
8	IS_{20} (%)	11.52 ±2.67	13.99 ±3.90	18.62 ±5.64	26.84 ±8.76	35.71 ±11.53
	OCR (%)	26.97 ±4.45	29.92 ±3.84	31.51 ±5.27	35.35 ±7.98	45.35 ±9.23
	LO (%)	83.58 ±4.07	84.95 ±3.07	82.51 ±4.03	80.80 ±5.25	81.14 ±4.23
10	IS_{20} (%)	8.83 ±3.15	9.74 ±4.54	14.27 ±5.61	21.46 ±8.88	31.21 ±11.56
	OCR (%)	33.62 ±4.18	34.65 ±4.93	37.23 ±5.53	40.62 ±8.71	52.59 ±9.17
	LO (%)	77.57 ±3.40	78.14 ±2.76	76.96 ±3.22	78.77 ±4.09	78.34 ±3.46
12	IS_{20} (%)	9.76 ±3.57	10.34 ±3.19	14.74 ±3.51	21.44 ±6.96	32.44 ±11.91
	OCR (%)	35.94 ±4.95	37.51 ±5.54	39.56 ±5.18	44.20 ±6.90	54.27 ±9.00
	LO (%)	80.69 ±2.98	80.95 ±3.40	80.61 ±2.78	81.46 ±3.27	80.77 ±4.11

minimization objectives. zgCircles represent redesigns and crosses frequency-only solutions. The third objective, structural continuity, is not shown in this projection; line overlap for the redesigned networks is reported in Table 1.



(a) Ahern et al. (b) Vermeir et al.

Figure 2: Trip Time Distribution Comparison Under Various Variability Conditions in a randomly chosen realization. zgPanels compare redesign against the static networks of Ahern et al. (2022) and Vermeir et al. (2021). Positive values indicate shorter trips under redesign.

For illustration, in this subsection we report the results for a single randomly chosen realization. zgFigure 2 shows the trip-time changes. At $cv = 1$, the plotted shares of travelers with shorter trips are 70.1% relative to Ahern et al. and 73.3% relative to Vermeir et al.; at $cv = 0.05$, they are 40.1% and 44.8%, respectively. These are illustrative single-realization results.

4.2 zg~~Large-scale validation on the Beijing network~~ Beijing network

We compare our method against a *stop-skipping* operational baseline on the Beijing network. For each variability level, links whose predicted travel time exceeds the original by more than zg~~a preset threshold~~a preset percentage are classified as degraded. The stop-skipping strategy then removes intermediate stops adjacent to degraded links whenever a direct shortcut is available, while leaving the overall line structure and vehicle allocation unchanged. zg~~The threshold thus controls aggressiveness: a 25% threshold skips stops whenever a link's predicted time grows by more than a quarter; a 100% threshold acts only when it doubles.~~ This baseline represents a simple, local response to travel-time disruption, but structurally

constrained: it cannot reroute vehicles or reassign fleet, and skipping stops may disconnect part of the demand at a stop from the network entirely, obliging users to walk to different stops.

Table 2: zgBeijing results at $cv = 1$ compared with the original network. Travel-time improvement for stop-skipping is calculated on reachable OD pairs.

Method	Overlap (%)	Cost reduction (%)	Travel-time improvement (%)	Unreachable OD (%)
Stop-skipping, 25% threshold	42.18	12.71	14.71	38.93
Stop-skipping, 50% threshold	47.83	12.39	19.71	35.28
Stop-skipping, 100% threshold	55.88	13.49	15.11	28.05
Redesign	85.42	22.56	25.78	zg~~–~~0

The results in Table 3 show that the original network exhibits some inefficiencies: the solution of our optimization yields an 11.40% zg~~Avg. TTI even at $cv = 0$~~average travel-time improvement even at $cv = 0$. zg~~This gain at $cv = 0$ does not stem from prediction-driven adaptation: with no travel-time variability, there is nothing to forecast or to adapt to.~~This reflects static re-optimisation of the incumbent network rather than prediction-driven adaptation. The benefit grows with variabilityzg~~: at higher cv, the algorithm produces designs increasingly different from the static original, achieving both travel-time improvements and line shortening, which in turn lowers operational cost~~. Table 2 exposes a basic tension in stop-skipping: operational benefit comes at the cost of connectivity. zg~~Aggressive thresholds improve travel time on the reachable network but disconnect a large share of OD pairs as variability grows, whereas conservative thresholds keep connectivity yet act too rarely to help and still lose connectivity at high variability.~~Aggressive thresholds improve travel time on the reachable network but disconnect a large share of OD pairs as variability grows. zg~~Because no stops are removed, all OD pairs stay reachable and the reported TTI covers the full demand, making the gains comparable across variability levels without the connectivity issues of every stop-skipping variant.~~Because no stops are removed, all OD pairs stay reachable under the redesign. zgTravel-time improvements for stop-skipping and redesign are calculated on different reachable OD sets and should therefore be interpreted alongside the connectivity figures.

Table 3: zgRedesign results on the Beijing network across traffic variability levels, compared with the original network.

Coefficient of variation	0	0.05	0.25	0.50	1.00
Line overlap (%)	90.08	87.34	86.51	86.39	85.42
Operational cost reduction (%)	0.50	1.53	3.85	12.02	22.56
Average travel-time improvement (%)	11.40	11.60	16.99	18.44	25.78

zgTable 3 shows the reported trade-off across all five settings. Average travel-time improvement is computed as the reduction in the sum of OD travel times relative to the original network, divided by the corresponding original sum; it is not the fraction of OD pairs improved by 20%, nor a demand-weighted average. Operational cost reduction reflects the model's vehicle-time proxy, which excludes costs such as crew scheduling and depot operations.

5 Conclusion

This paper systematically studied the potential of proactive bus line redesign, targeted at limited parts of the network, in maintaining user and operator performance in the case of severe traffic fluctuations. We formulated a multi-objective optimization problem balancing passenger travel time, operational costs, and network stability. Our predict-then-optimize approach relies on an NSGA-III-based optimization framework that proactively redesigns bus lines based on predicted line travel times.zg ~~We evaluated our approach against the inevitable prediction errors of well-established traffic-predicting models. To this aim, we expose the optimizer to realistic forecast inaccuracies sampled from a well-established deep learning-based prediction model (DCRNN).~~ Experiments on Mandl's network and a real-world

Beijing network demonstrate effective adaptation across varying levels of travel-time uncertainty.zg ~~Results indicate that, faced with non-negligible link travel-time variability, the proposed approach yields OD-level trip travel-time reductions, while achieving operational cost savings over the selected baselines and scenarios considered, and preserving a high degree of structural similarity with the baseline network.~~ zg~~These findings highlight the importance of moving beyond static network design toward continuous, adaptive redesign in dynamic operating environments.~~Overall, the gains grow with travel-time variability and come with high line overlap, indicating that the benefit does not rely on large structural change.

A natural extension is to enrich the forecast-error model with explicit spatial and temporal correlation, allowing network-wide common shocks to be represented. zg~~A further direction is to incorporate in-vehicle crowding, for instance through a capacitated or congestion-dependent (frequency-based common-line) passenger assignment; this would let the redesign account for the load it induces on each line and report crowding-aware rather than purely uncongested travel-time gains.~~A further direction is to incorporate in-vehicle crowding, for instance through a capacitated or congestion-dependent passenger assignment; this would let the redesign account for the load it induces on each line and report crowding-aware rather than purely uncongested travel-time gains.

zgDeployment also requires advance notice, passenger information, and operator coordination; these practical costs and passenger responses are not evaluated here.

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